

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Percentages

02

10 answers · Rationale-rich · Teach from doc

Q1**203/50**

Accept also: 4.06

The correct answer is 4.06. It's given that the retail price is 290% of the wholesale price of \$ 7.00. Thus, the retail price is $\$ 7.00 \frac{290}{100}$, which is equivalent to \$ 7.002.9, or \$ 20.30. It's also given that the discounted price is 80% off the retail price. Thus, the discounted price is $\$ 20.301 - \frac{80}{100}$, which is equivalent to \$ 20.300.20, or \$ 4.06.

Q2**600**

The correct answer is 600. It's given that 210 is $p\%$ greater than 30. It follows that $210 = 1 + \frac{p}{100}30$. Dividing both sides of this equation by 30 yields $7 = 1 + \frac{p}{100}$. Subtracting 1 from both sides of this equation yields $6 = \frac{p}{100}$. Multiplying both sides of this equation by 100 yields $p = 600$. Therefore, the value of p is 600.

Q3**D**

D. 4,941%

Choice D is correct. It's given that a is 2,241% of the sum of b and c . This can be represented by the equation $a = \frac{2,241}{100}b + c$, or $a = 22.41b + c$. It's also given that b is 83% of c . This can be represented by the equation $b = \frac{83}{100}c$, or $b = 0.83c$. Dividing both sides of this equation by 0.83 yields $\frac{b}{0.83} = c$. Substituting $\frac{b}{0.83}$ for c in the equation $a = 22.41b + c$ yields $a = 22.41b + \frac{b}{0.83}$, or $a = 22.41\frac{1.83b}{0.83}$, which is equivalent to $a = \frac{41.0103b}{0.83}$, or $a = 49.41b$. This equation is equivalent to $a = \frac{4,941}{100}b$; therefore, a is 4,941% of b .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4**A****A. 12**

Choice A is correct. It's given that the result of increasing the quantity x by 400% is 60. This can be written as $x + \frac{400}{100}x = 60$, which is equivalent to $x + 4x = 60$, or $5x = 60$. Dividing each side of this equation by 5 yields $x = 12$. Therefore, the value of x is 12.

Choice B is incorrect. The result of increasing the quantity 15 by 400% is 75, not 60.

Choice C is incorrect. The result of increasing the quantity 240 by 400% is 1,200, not 60.

Choice D is incorrect. The result of increasing the quantity 340 by 400% is 1,700, not 60.

Q5**79**

The correct answer is 79. Let x represent the number of zebras in the population in 2014 and let y represent the number of zebras in the population in 2018. It's given that the number of zebras in this population in 2018 was 1.27 times the number of zebras in this population in 2014. It follows that the equation $y = 1.27x$ represents this situation. Dividing both sides of this equation by 1.27 yields $\frac{y}{1.27} = x$, or $\frac{1}{1.27}y = x$. Therefore, the number of zebras in this population in 2014 is $\frac{1}{1.27}$ times the number of zebras in this population in 2018. If the number of zebras in this population in 2014 is $p\%$ of the number of zebras in this population in 2018, then $x = \frac{p}{100}y$. It follows that $\frac{1}{1.27} = \frac{p}{100}$, or $\frac{100}{1.27} = p$, which means p is approximately equal to 78.74. Therefore, the value of p , to the nearest whole number, is 79.

Q6**1512/5**

Accept also: 302.4

The correct answer is 302.4. It's given that the number b is 320% greater than 160. It follows that b is equal to 160 plus 320% of 160, which can be written as $b = 160 + 3.2(160)$, or $b = 672$. It's also given that the number a is 55% less than the number b . It follows that a is equal to b minus 55% of b , which can be written as $a = b - 0.55b$, or $a = 0.45b$. Substituting 672 for b in this equation yields $a = 0.45(672)$, or $a = 302.4$.

Q7**4.05**

Accept also: 81/20

The correct answer is 4.05. It's given that the purchase price for certain books is 5.00 dollars each. It's also given that each book is marked for sale at a consumer price that is 270% of the purchase price. Since the consumer price is 270% of the purchase price of 5.00 dollars, it follows that the consumer price is $(2.7)(5.00)$, or 13.50, dollars. It's given that after 4 months, any remaining books are discounted at 70% off the consumer price. Thus, the discount amount is $(0.7)(13.50)$, or 9.45, dollars. Subtracting the discount amount from the consumer price gives the discounted price of each of the remaining books: $13.50 - 9.45 = 4.05$. Note that 4.05 and 81/20 are examples of ways to enter a correct answer.

Q8**8**

The correct answer is 8. The 6 students represent $(100 - 15 - 45 - 25)\% = 15\%$ of those invited to join the committee. If x people were invited to join the committee, then $0.15x = 6$. Thus, there were $\frac{6}{0.15} = 40$ people invited to join the committee. It follows that there were $0.45(40) = 18$ teachers and $0.25(40) = 10$ school and district administrators invited to join the committee. Therefore, there were 8 more teachers than school and district administrators invited to join the committee.

Q9**.54**

Accept also: 27/50

The correct answer is .54. It's given that the number a is 70% less than the positive number b . Therefore, $a = 1 - \frac{70}{100}b$, which is equivalent to $a = 1 - 0.70b$, or $a = 0.30b$. It's also given that the number c is 80% greater than a . Therefore, $c = 1 + \frac{80}{100}a$, which is equivalent to $c = 1 + 0.80a$, or $c = 1.80a$. Since $a = 0.30b$, substituting $0.30b$ for a in the equation $c = 1.80a$ yields $c = 1.80(0.30b)$, or $c = 0.54b$. Thus, c is 0.54 times b . Note that .54 and 27/50 are examples of ways to enter a correct answer.

Q10

60000

The correct answer is 60,000. It's given that the weight of sample A is 468% of the weight of sample B and 0.780% of the weight of sample C. Let the weight of sample B be b and the weight of sample C be c . Then, the weight of sample A is equivalent to $\frac{468b}{100}$, or $4.68b$, and also equivalent to $\frac{0.780c}{100}$, or $0.00780c$. Setting these expressions equal to each other yields $4.68b = 0.00780c$. Dividing both sides of this equation by 0.00780 yields $\frac{4.68b}{0.00780} = c$, or $c = 600b$. If the weight of sample C is $p\%$ of the weight of sample B, then $c = \frac{p}{100}b$. Since $c = 600b$ and $c = \frac{p}{100}b$, it follows that $600b = \frac{p}{100}b$, or $600 = \frac{p}{100}$. Multiplying both sides of this equation by 100 yields $60,000 = p$. Therefore, the value of p is 60,000.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank